

1.8 Proof Methods and Strategy

1.8.2 Exhaustive Proof and Proof by Cases

Sometimes we cannot prove a theorem using a single argument that holds for all possible cases. We now introduce a method that can be used to prove a theorem by considering different cases separately.

This method is based on a rule of inference that we will now introduce. To prove a conditional statement of the form $(P_1 \vee P_2 \vee \dots \vee P_n) \rightarrow Q$, the tautology $[(P_1 \vee P_2 \vee \dots \vee P_n) \rightarrow Q] \leftrightarrow [(P_1 \rightarrow Q) \wedge (P_2 \rightarrow Q) \wedge \dots \wedge (P_n \rightarrow Q)]$ can be used as a rule of inference. This shows that the original conditional statement with a hypothesis made up of the disjunction of the propositions $P_1, P_2, \dots, P_n$ can be proved by proving each of the n conditional statements $P_i \rightarrow Q, i=1, 2, \dots, n$ individually. Such an argument is called a proof by cases.

Sometimes, to prove that a conditional statement $(P \rightarrow Q)$ is True, it is convenient to use a disjunction $P_1 \vee P_2 \vee \dots \vee P_n$ instead of P as the hypothesis of the conditional statement, where P and $P_1 \vee P_2 \vee \dots \vee P_n$ are equivalent.